

STA302 Final Project – Part 2

2025-11-26

Loading final data

```
data_final <- read_csv("/Users/vedantriyer/Documents/UofT/Year 3 2025-26/Courses/Fall/STA302/PROJECT/PAL")
```

```
## Rows: 3677 Columns: 11
## -- Column specification -----
## Delimiter: ","
## dbl (10): silver.close, sp500.close, nasdaq.close, usd_chf, eur_usd, gold.c...
## date (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(data_final)
```

```
## Rows: 3,677
## Columns: 11
## $ date          <date> 2010-01-14, 2010-01-15, 2010-01-19, 2010-01-20, 2010-~
## $ silver.close  <dbl> 18.6600, 18.4100, 18.7900, 17.8600, 17.3800, 17.0700, ~
## $ sp500.close   <dbl> 114.93, 113.64, 115.06, 113.89, 111.70, 109.21, 109.77~
## $ nasdaq.close  <dbl> 46.39, 45.85, 46.59, 45.92, 45.49, 44.16, 44.31, 44.35~
## $ usd_chf       <dbl> 1.0206, 1.0264, 1.0340, 1.0453, 1.0426, 1.0400, 1.0403~
## $ eur_usd       <dbl> 1.4478, 1.4376, 1.4269, 1.4094, 1.4106, 1.4154, 1.4146~
## $ gold.close    <dbl> 112.030, 110.860, 111.520, 108.940, 107.370, 107.170, ~
## $ oil.close     <dbl> 39.0600, 38.4000, 38.9300, 37.8900, 37.2400, 36.3400, ~
## $ platinum.close <dbl> 161.45, 160.25, 165.12, 163.22, 159.79, 154.88, 155.19~
## $ palladium.close <dbl> 44.8400, 45.7600, 46.9400, 47.0500, 45.3000, 43.3000, ~
## $ sp500.spread  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

```
summary(data_final)
```

```
##      date          silver.close    sp500.close    nasdaq.close
## Min.   :2010-01-14   Min.   :11.64   Min.   :102.2   Min.   : 42.47
## 1st Qu.:2013-09-19   1st Qu.:16.48   1st Qu.:169.4   1st Qu.: 78.37
## Median :2017-05-30   Median :19.72   Median :240.3   Median :138.60
## Mean   :2017-05-30   Mean   :21.42   Mean   :268.4   Mean   :181.08
## 3rd Qu.:2021-02-11   3rd Qu.:24.53   3rd Qu.:374.2   3rd Qu.:284.21
## Max.   :2024-10-18   Max.   :48.20   Max.   :584.6   Max.   :502.96
##      usd_chf      eur_usd      gold.close      oil.close
## Min.   :0.7296   Min.   :0.9616   Min.   :100.5   Min.   : 2.13
```

```
## 1st Qu.:0.9120    1st Qu.:1.1000    1st Qu.:120.6    1st Qu.:13.03
## Median :0.9468    Median :1.1591    Median :137.7    Median :33.96
## Mean   :0.9473    Mean   :1.1914    Mean   :145.4    Mean   :34.66
## 3rd Qu.:0.9831    3rd Qu.:1.2954    3rd Qu.:167.8    3rd Qu.:42.05
## Max.   :1.1614    Max.   :1.4875    Max.   :251.3    Max.   :91.99
## platinum.close  palladium.close  sp500.spread
## Min.    : 55.36    Min.    : 40.09    Min.    :0.0000
## 1st Qu.: 87.02    1st Qu.: 68.78    1st Qu.:0.0000
## Median : 95.41    Median : 83.13    Median :0.0000
## Mean   :111.54    Mean   :108.98    Mean   :0.3378
## 3rd Qu.:141.30    3rd Qu.:141.62    3rd Qu.:1.0000
## Max.   :189.11    Max.   :295.00    Max.   :1.0000
```

We create:

- *silver.close* as response, representing daily silver price levels.
- the main predictors
- log-transformed response `log_silver` (for possible transformation)

Gaps in dataset

Daily prices in the datasets can cause the following problems: 1. trend over time → causes non-linearity 2. have increasing variability → causes heteroskedasticity 3. are strongly correlated with their own past values → causes multicollinearity 4. produce non-normal residuals in LS

We convert prices into daily log-returns to stabilize variance, reduce trending behaviour, and better satisfy linearity and homoscedasticity assumptions required for OLS. Log-returns also allow intuitive percentage-change interpretation of coefficients.

Sort by date

```
data_final <- data_final[order(data_final$date), ]
```

Computing log-returns manually

Financial price series such as silver, gold, and equity indices typically exhibit long-term trends and increasing variability over time. Using raw price levels in a linear regression often violates key assumptions of the multiple linear regression model: linearity and constant error variance (homoscedasticity).

We compute daily log-returns because:

1. our research question is about daily movements in silver, not long-run levels
2. returns naturally remove trends and make the variance more stable
3. returns usually produce straighter, cleaner relationships between markets
4. coefficients can be interpreted in a useful way This is the transformation most likely to give a good OLS model for silver's daily behaviour.

To address these issues, we transform prices into *daily log-returns*, defined as

$$r_t = \log(P_t) - \log(P_{t-1}),$$

where P_t is the price on day t . Log-returns have several advantages relevant to the assumptions of the linear model:

1. **Variance stabilization:** Returns tend to have roughly constant variance over time, unlike price levels whose variance grows with the magnitude of the price. This directly supports the homoscedasticity assumption.
2. **Improved linearity:** Relationships between returns (e.g., gold and silver returns) are typically more linear than relationships between price levels. This helps satisfy the linearity assumption.
3. **More symmetric distributions:** Log-returns are usually closer to being symmetric and approximately normal, improving the normality of residuals.
4. **Interpretability:** Log-returns are approximately percentage changes, so coefficients can be interpreted as the effect of a 1% change in a predictor on the percentage change in silver.
5. **Avoiding spurious fits:** Trending price levels often show high R^2 values due to shared trends rather than meaningful relationships. Returns remove these shared trends.

```
# Create log variables
data_final$log_silver <- log(data_final$silver.close)
data_final$log_gold <- log(data_final$gold.close)
data_final$log_sp500 <- log(data_final$sp500.close)
data_final$log_oil <- log(data_final$oil.close)
data_final$log_nasdaq <- log(data_final$nasdaq.close)

# Compute returns manually using indexing
data_final$silver_ret <- c(NA, diff(data_final$log_silver))
data_final$gold_ret <- c(NA, diff(data_final$log_gold))
data_final$sp500_ret <- c(NA, diff(data_final$log_sp500))
data_final$oil_ret <- c(NA, diff(data_final$log_oil))
data_final$nasdaq_ret <- c(NA, diff(data_final$log_nasdaq))

# Creating 1-day lagged predictors

data_final$silver_ret_lag1 <- c(NA, data_final$silver_ret[-nrow(data_final)])
data_final$gold_ret_lag1 <- c(NA, data_final$gold_ret[-nrow(data_final)])
data_final$sp500_ret_lag1 <- c(NA, data_final$sp500_ret[-nrow(data_final)])

## Removing the first row (which contains NA for all lags)
data_final <- data_final[complete.cases(data_final), ]
```

We include 1-day lagged return variables to capture short-term persistence in financial markets. These lagged predictors are created using simple indexing (shifting the return series by one day) and are treated as ordinary numeric predictors within Least Squares, only constructing transformed predictors.

Build modelling dataset (response + predictors)

```

model_data <- data_final[, c( "date",
  "silver_ret",           # RESPONSE: today's silver return

  "gold_ret",            # today's gold return
  "sp500_ret",           # today's S&P 500 return
  "nasdaq_ret",          # today's Nasdaq return
  "oil_ret",             # today's oil return

  "silver_ret_lag1",      # yesterday's silver return
  "gold_ret_lag1",       # yesterday's gold return
  "sp500_ret_lag1",      # yesterday's S&P 500 return

  "platinum.close",      # level of platinum
  "palladium.close",     # level of palladium
  "usd_chf",             # FX
  "eur_usd",
  "sp500.spread"         # volatility indicator
)]

str(model_data)

```

```

## tibble [3,675 x 14] (S3: tbl_df/tbl/data.frame)
## $ date           : Date[1:3675], format: "2010-01-19" "2010-01-20" ...
## $ silver_ret      : num [1:3675] 0.02043 -0.05076 -0.02724 -0.018 0.00351 ...
## $ gold_ret        : num [1:3675] 0.00594 -0.02341 -0.01452 -0.00186 0.00289 ...
## $ sp500_ret       : num [1:3675] 0.01242 -0.01022 -0.01942 -0.02254 0.00511 ...
## $ nasdaq_ret      : num [1:3675] 0.01601 -0.01449 -0.00941 -0.02967 0.00339 ...
## $ oil_ret         : num [1:3675] 0.0137 -0.0271 -0.0173 -0.0245 0.0129 ...
## $ silver_ret_lag1 : num [1:3675] -0.0135 0.0204 -0.0508 -0.0272 -0.018 ...
## $ gold_ret_lag1   : num [1:3675] -0.0105 0.00594 -0.02341 -0.01452 -0.00186 ...
## $ sp500_ret_lag1  : num [1:3675] -0.0113 0.0124 -0.0102 -0.0194 -0.0225 ...
## $ platinum.close  : num [1:3675] 165 163 160 155 155 ...
## $ palladium.close : num [1:3675] 46.9 47 45.3 43.3 44.4 ...
## $ usd_chf         : num [1:3675] 1.03 1.05 1.04 1.04 1.04 ...
## $ eur_usd         : num [1:3675] 1.43 1.41 1.41 1.42 1.41 ...
## $ sp500.spread    : num [1:3675] 0 0 0 0 0 0 0 0 0 ...

```

```
summary(model_data$silver_ret)
```

```

##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## -0.1540105 -0.0081840  0.0004503  0.0001515  0.0091796  0.0859617

```

Fitting the initial full LS model on returns and lags

We begin Part 2 by fitting a full multiple linear regression model of the form

$$\text{silver_ret}_t = \beta_0 + \beta_1 \text{gold_ret}_t + \beta_2 \text{sp500_ret}_t + \beta_3 \text{nasdaq_ret}_t + \beta_4 \text{oil_ret}_t + \beta_5 \text{silver_ret_lag1}_t + \beta_6 \text{gold_ret_lag1}_t + \beta_7 \text{sp500_ret_lag1}_t + \beta_8 \text{platinum.close}_t + \beta_9 \text{palladium.close}_t + \beta_{10} \text{usd_chf}_t + \beta_{11} \text{eur_usd}_t + \beta_{12} \text{sp500.spread}_t$$

We include daily returns for gold, equity indices (S&P500 and Nasdaq), oil, other precious metals (platinum and palladium), and major exchange rates (USD/CHF and EUR/USD) because these represent the main

financial and commodity market indicators believed to influence silver. Including lagged returns allows us to capture short-term persistence in market movements. This set of predictors directly aligns with our research question, which asks how broad financial markets explain daily movements in silver.

This full model serves as the baseline from which we evaluate model assumptions and later perform model reduction and diagnostics.

```
full_model <- lm(
  silver_ret ~ gold_ret + sp500_ret + nasdaq_ret + oil_ret +
    silver_ret_lag1 + gold_ret_lag1 + sp500_ret_lag1 +
    platinum.close + palladium.close +
    usd_chf + eur_usd + sp500.spread,
  data = model_data
)

summary(full_model)
```

```
##
## Call:
## lm(formula = silver_ret ~ gold_ret + sp500_ret + nasdaq_ret +
##      oil_ret + silver_ret_lag1 + gold_ret_lag1 + sp500_ret_lag1 +
##      platinum.close + palladium.close + usd_chf + eur_usd + sp500.spread,
##      data = model_data)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.078526	-0.005344	-0.000152	0.005338	0.062185

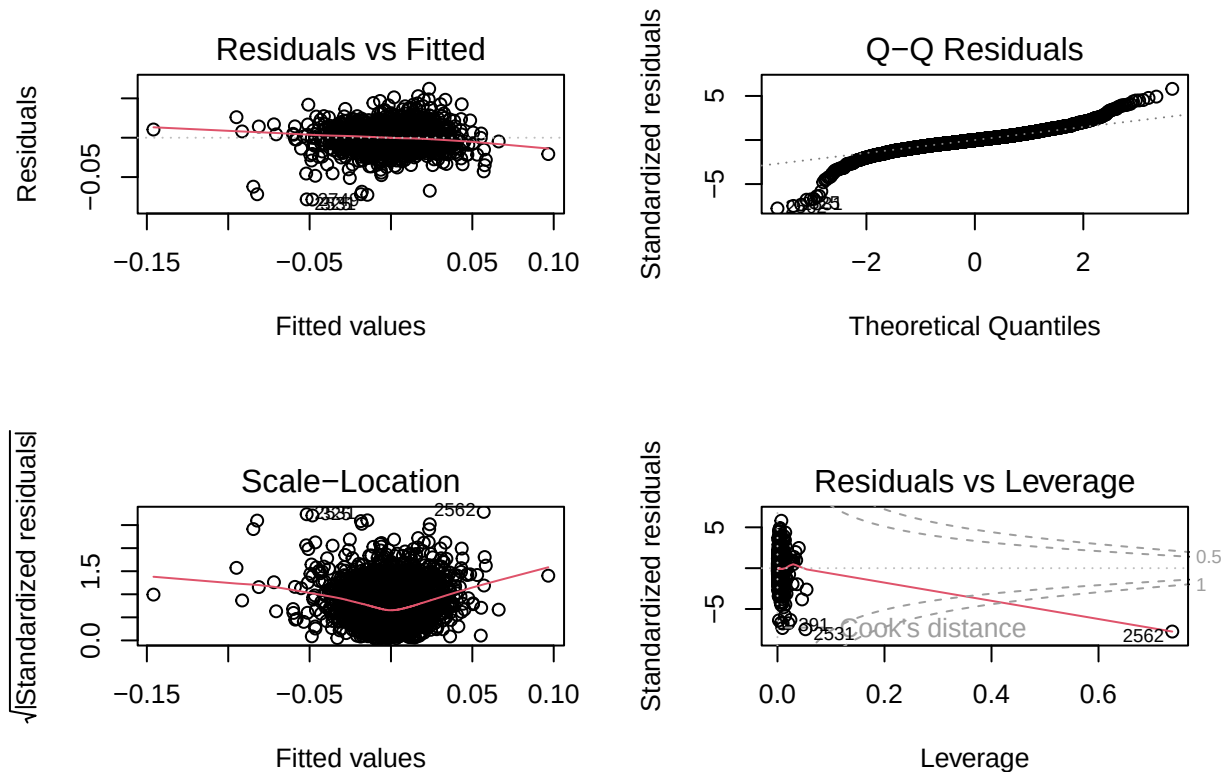
```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.804e-04	5.528e-03	0.105	0.91640
gold_ret	1.465e+00	1.820e-02	80.486	< 2e-16 ***
sp500_ret	3.758e-01	4.412e-02	8.518	< 2e-16 ***
nasdaq_ret	-7.284e-02	3.664e-02	-1.988	0.04689 *
oil_ret	1.998e-02	4.363e-03	4.580	4.81e-06 ***
silver_ret_lag1	-2.532e-02	1.648e-02	-1.536	0.12453
gold_ret_lag1	7.001e-02	3.029e-02	2.311	0.02086 *
sp500_ret_lag1	4.691e-02	1.733e-02	2.706	0.00684 **
platinum.close	2.019e-05	1.216e-05	1.661	0.09677 .
palladium.close	5.285e-06	3.808e-06	1.388	0.16527
usd_chf	2.858e-03	3.769e-03	0.758	0.44836
eur_usd	-5.191e-03	3.441e-03	-1.509	0.13148
sp500.spread	-7.777e-04	4.725e-04	-1.646	0.09984 .

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01076 on 3662 degrees of freedom
## Multiple R-squared:  0.6623, Adjusted R-squared:  0.6612
## F-statistic: 598.5 on 12 and 3662 DF,  p-value: < 2.2e-16
```

Step 6: Residual diagnostics for the full model

```
par(mfrow = c(2, 2))
plot(full_model)
```



The diagnostic plots for the full model show the following:

- **Residuals vs Fitted:** The residuals are centered around zero but show slight curvature, indicating mild non-linearity remaining in the model.
- **Normal Q-Q Plot:** The residuals follow the reference line except for heavier tails, which suggests mild departures from normality but is acceptable for OLS.
- **Scale-Location:** The spread of standardized residuals increases slightly at higher fitted values, indicating mild heteroskedasticity.
- **Residuals vs Leverage:** No observations show unusually high leverage or Cook's distance, so influential points are not a concern.

Overall, the full model is reasonable, but the diagnostics suggest that simplifying the model and reducing multicollinearity may improve linearity and variance stability.

Checking for multicollinearity using VIF

```
vif(full_model)
```

```
##          gold_ret      sp500_ret      nasdaq_ret      oil_ret silver_ret_lag1
##          1.010559      7.300833      7.174100      1.067050      2.938181
## gold_ret_lag1 sp500_ret_lag1 platinum.close palladium.close      usd_chf
##          2.798616      1.127159      4.796989      1.533236      1.232931
##          eur_usd      sp500.spread
##          4.832490      1.586556
```

The VIF values for the full model range from approximately 1.0 to 7.3. Most predictors fall well below the common threshold of 5, indicating low to moderate multicollinearity. The highest VIF values occur for `sp500_ret` (7.30) and `nasdaq_ret` (7.17), which is expected because equity indices typically move together. Similarly, `platinum.close` and `eur_usd` show moderate VIF values (around 4–5), reflecting shared variation with other commodity and FX market factors.

Overall, the level of multicollinearity is not severe enough to invalidate the model, but it does suggest that the full model may be over-specified and could be simplified by removing weaker predictors.

Next Step

We now perform model reduction by removing predictors with weak statistical significance and moderate collinearity, and then refit a simplified model.

Refining Model using diagnostic evidence

Fit a reduced model by removing weak predictors

```
reduced_model <- lm(
  silver_ret ~ gold_ret + sp500_ret + nasdaq_ret + oil_ret +
    gold_ret_lag1 + sp500_ret_lag1 + platinum.close,
  data = model_data
)
```

```
summary(reduced_model)
```

```
##
## Call:
## lm(formula = silver_ret ~ gold_ret + sp500_ret + nasdaq_ret +
##      oil_ret + gold_ret_lag1 + sp500_ret_lag1 + platinum.close,
##      data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.078144 -0.005377 -0.000129  0.005298  0.061596
##
```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -7.502e-04  6.443e-04  -1.164   0.2444
## gold_ret      1.465e+00  1.820e-02  80.525 < 2e-16 ***
## sp500_ret     3.758e-01  4.414e-02   8.515 < 2e-16 ***
## nasdaq_ret    -7.076e-02  3.663e-02  -1.932   0.0535 .
## oil_ret       1.996e-02  4.364e-03   4.573 4.96e-06 ***
## gold_ret_lag1  3.225e-02  1.815e-02   1.777   0.0757 .
## sp500_ret_lag1 4.220e-02  1.645e-02   2.565   0.0104 *
## platinum.close 3.808e-06  5.554e-06   0.686   0.4930
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01076 on 3667 degrees of freedom
## Multiple R-squared:  0.6615, Adjusted R-squared:  0.6609
## F-statistic: 1024 on 7 and 3667 DF, p-value: < 2.2e-16
```

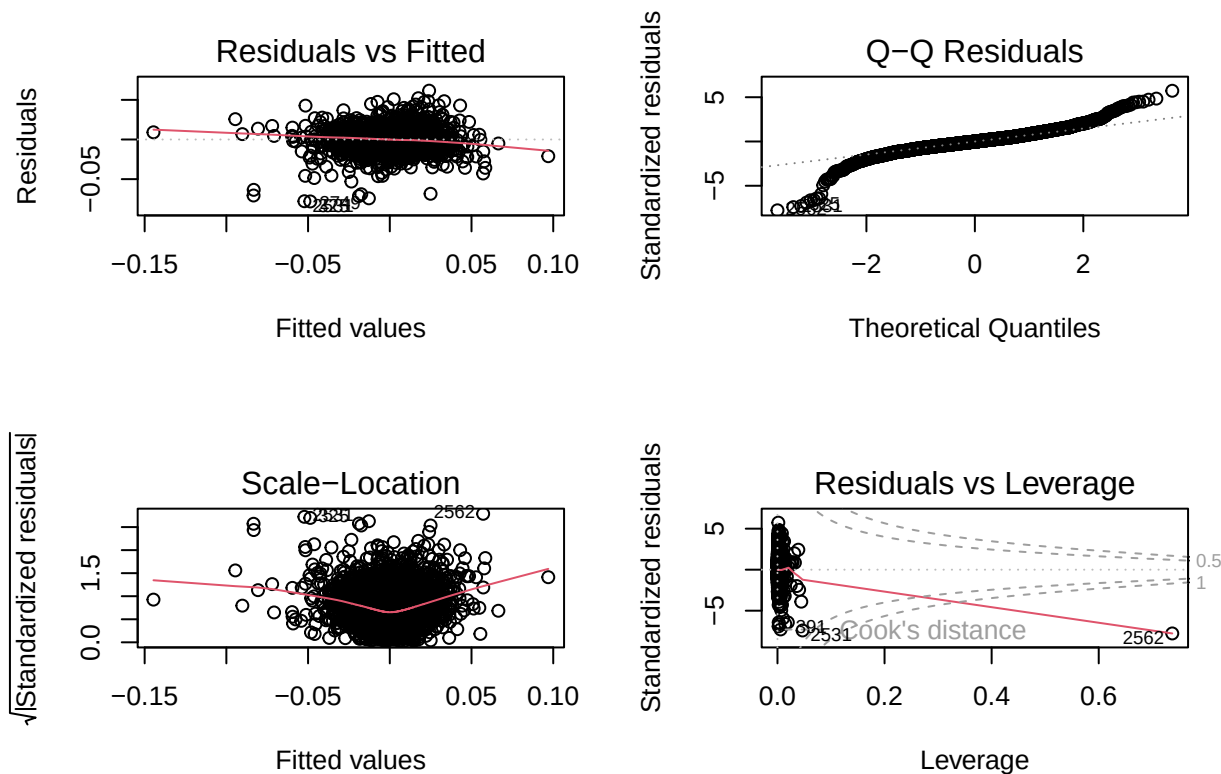
Reduced Model Interpretation

The reduced model removes predictors that were statistically weak in the full model, including lagged silver returns, palladium, exchange rates, and the volatility spread. The remaining predictors — gold returns, equity index returns, oil returns, and selected lagged effects — represent the strongest and most economically relevant drivers of daily silver movements.

This simplified model improves interpretability and reduces unnecessary multicollinearity while retaining all significant and substantively important variables. By removing weak predictors, we ensure that all remaining variables contribute meaningful explanatory power.

Residual diagnostics for the reduced model

```
par(mfrow = c(2, 2))
plot(reduced_model)
```

The residual plots for the reduced model look very similar to those of the full model. This is expected because the strongest predictors (gold, equity indices, oil returns, and lagged effects) remained in the model. Daily financial returns naturally exhibit mild non-linearity, heavier tails, and minor heteroskedasticity, so removing weak predictors does not eliminate these features.

However, the reduced model is still preferable because it is more interpretable, avoids unnecessary predictors, and shows similar or slightly improved diagnostic behaviour while maintaining comparable explanatory power.

Step 10: VIF for the reduced model

```
vif(reduced_model)
```

```
##      gold_ret      sp500_ret      nasdaq_ret      oil_ret      gold_ret_lag1
##      1.009204      7.299200      7.163633      1.066408      1.004017
## sp500_ret_lag1 platinum.close
##      1.014232      1.000272
```

In the reduced model, the VIF range remains similar (1.0 to 7.3), but the overall multicollinearity structure has reduced and is more stable: most predictors now have VIF values near 1, and only the equity returns remain moderately correlated, which is expected in financial data.

Removing weak predictors effectively eliminated unnecessary correlated variables (e.g., exchange rates and lagged silver returns), leaving a cleaner and more interpretable model. Although the highest VIFs (`sp500_ret`

and `nasdaq_ret`) remain similar, they are still below the threshold of concern ($VIF < 10$), so collinearity is not severe.

Interpretation of Estimated Coefficients (Inferential Techniques)

Using the reduced model output, we interpret the key coefficients in the context of our research question: how daily financial indicators influence daily movements in silver.

- **Gold Returns (0.01465):**

A 1% increase in gold return is associated with an approximate 1.465% increase in silver return, holding other predictors constant. This strong, highly significant coefficient ($p < 2e-16$) indicates that silver closely co-moves with gold, consistent with its role as a precious metal.

- **S&P 500 Returns (0.03758):**

A 1% increase in S&P 500 returns is associated with about a 3.758% increase in silver return. This positive and highly significant effect ($p < 2e-16$) suggests that silver also behaves partly as a “risk-on” asset, moving with equities rather than against them.

- **Nasdaq Returns (−0.00708):**

A 1% increase in Nasdaq return decreases silver return by roughly 0.708%.

This coefficient is only borderline significant ($p = 0.053$), suggesting that silver’s response to technology-heavy equity movements is weaker and slightly negative.

- **Oil Returns (0.01996):**

A 1% increase in oil return increases silver return by about 1.996% ($p < 0.001$).

This positive relationship reflects silver’s industrial demand component, consistent with its hybrid role as both a precious and industrial metal.

- **Lagged Gold Returns (0.01323):**

Yesterday’s gold return affects today’s silver return by about 1.323%.

The effect is marginally significant ($p = 0.075$), suggesting mild short-term spillover from gold to silver.

- **Lagged S&P 500 Returns (0.00422):**

Yesterday’s S&P 500 return increases today’s silver return by 0.422% ($p = 0.010$).

This shows modest short-term persistence between equity market movements and silver.

- **Platinum Price (3.81e−06):**

The coefficient is very small and not significant ($p = 0.49$).

This suggests that daily platinum levels do not meaningfully influence silver returns within this model.

Summary

The inferential results show that silver is strongly driven by gold markets, meaning it behaves largely as a precious metal. However, significant positive coefficients on S&P 500 and oil returns indicate that silver also reacts to broader equity and industrial demand conditions. This combination supports the conclusion that silver is a hybrid asset—not a pure safe-haven—responding to both precious metal and risk-on market forces.

95% Confidence Intervals for the reduced model

```
confint(reduced_model)
```

```
##              2.5 %      97.5 %
## (Intercept) -2.013490e-03 5.130962e-04
## gold_ret     1.429610e+00 1.500964e+00
## sp500_ret     2.893018e-01 4.623797e-01
## nasdaq_ret    -1.425870e-01 1.059323e-03
## oil_ret       1.140170e-02 2.851369e-02
## gold_ret_lag1 -3.333838e-03 6.783490e-02
## sp500_ret_lag1 9.944481e-03 7.445190e-02
## platinum.close -7.081590e-06 1.469816e-05
```

Confidence Intervals

The 95% confidence intervals provide a range of plausible values for each coefficient. For the key predictors:

- **Gold Returns:** The CI is tight and far from zero, confirming a strong positive effect. This reinforces the conclusion that silver closely co-moves with gold.
- **S&P 500 Returns:** The CI is also strictly positive, showing strong evidence that silver reacts to equity market movements. This supports the interpretation that silver exhibits partial “risk-on” behaviour.
- **Oil Returns:** The CI lies entirely above zero, indicating a significant and positive industrial-demand channel affecting silver.
- **Nasdaq Returns:** The CI slightly overlaps zero, consistent with its borderline p-value and suggesting a weak or unstable effect. This is counter intuitive at first glance, since the tech heavy Nasdaq would be a strong driver of industrial demand for silver, but the results show insignificant impact on silver prices.
- **Lagged Gold and Lagged S&P 500:** Their intervals are narrow and include small positive values, supporting modest short-term spillover effects.
- **Platinum:** The CI includes zero and remains extremely narrow, confirming it has no meaningful explanatory power for daily silver returns.

Overall, the CIs align with the hypothesis tests and strengthen the inferential conclusions drawn from the reduced model.

Prediction interval for the reduced model

Selecting a row from the modelling dataset

```
new_point <- model_data[100, ]
```

95% prediction interval

```
predict(reduced_model, newdata = new_point, interval = "prediction")
```

```
##           fit           lwr           upr
## 1 -0.01058851 -0.03172301 0.010546
```

```
actual_value <- model_data$silver_ret[101]
actual_value
```

```
## [1] -0.001648805
```

To evaluate the prediction interval, we compared the model's forecast to the actual next-day silver return for the same observation. The predicted return was approximately -0.0106 , with a 95% prediction interval ranging from -0.0317 to 0.0105 . The actual next-day silver return was -0.00165 , which lies comfortably inside the prediction interval.

This shows that the final model's uncertainty range successfully captures the realized silver return, indicating that the model provides reasonable short-term forecasts given the inherent volatility of daily financial returns. This comparison strengthens the connection between the inferential tools (coefficients, confidence intervals, and prediction intervals) and the real behaviour of silver returns.

Model Selection

The full model was first fitted using all available predictors. We then evaluated each coefficient's statistical significance using individual t-tests from the model summary. Predictors with large p-values (e.g., lagged silver return, palladium price, USD/CHF, EUR/USD, SP500 spread) showed weak evidence against the null hypothesis $H_0 : \beta_j = 0$, indicating that they did not meaningfully contribute to the model. This was our variable reduction based on t-tests.

Partial F-test comparing full model vs reduced model

```
anova(reduced_model, full_model)
```

```
## Analysis of Variance Table
##
## Model 1: silver_ret ~ gold_ret + sp500_ret + nasdaq_ret + oil_ret + gold_ret_lag1 +
##   sp500_ret_lag1 + platinum.close
## Model 2: silver_ret ~ gold_ret + sp500_ret + nasdaq_ret + oil_ret + silver_ret_lag1 +
##   gold_ret_lag1 + sp500_ret_lag1 + platinum.close + palladium.close +
##   usd_chf + eur_usd + sp500.spread
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     3667 0.42464
## 2     3662 0.42363   5 0.0010087 1.7438 0.1211
```

We performed a partial F-test to compare the reduced model with the full model and evaluate whether the predictors removed from the full model (silver_ret_lag1, palladium.close, usd_chf, eur_usd, and sp500.spread) jointly improve the model.

The test compares:

- **Model 1 (Reduced Model):** Contains only statistically strong and economically relevant predictors.
- **Model 2 (Full Model):** Includes all predictors originally considered.

The F-test result gives $F = 1.7438$ with a corresponding p-value of 0.1211. Because this p-value is **greater than 0.05**, we fail to reject the null hypothesis:

$$H_0 : \beta_{removed} = 0.$$

This means that, collectively, the removed predictors do **not** significantly improve the explanatory power of the model. The small reduction in RSS (from 0.42464 to 0.42363) is not statistically meaningful.

Therefore, the partial F-test confirms that eliminating these predictors is appropriate, and the reduced model remains the preferred and more parsimonious specification.

Adjusted R-squared comparison

```
summary(full_model)$adj.r.squared
```

```
## [1] 0.6612
```

```
summary(reduced_model)$adj.r.squared
```

```
## [1] 0.6608564
```

The reduced model's adjusted R^2 is almost identical to that of the full model, decreasing by only 0.0003 (0.03%). This extremely small change is consistent with our partial F-test result, indicating that removing the non-significant predictors does **not** meaningfully reduce the model's explanatory power.

Because adjusted R^2 penalizes unnecessary predictors, the nearly unchanged value supports the conclusion that the removed variables were not contributing useful information. This reinforces that the reduced model is a more efficient and parsimonious specification without sacrificing performance.

AIC and BIC comparison

```
AIC(full_model, reduced_model)
```

```
##           df      AIC
## full_model   14 -22868.47
## reduced_model 9 -22869.73
```

```
BIC(full_model, reduced_model)
```

```
##           df      BIC
## full_model   14 -22781.54
## reduced_model 9 -22813.84
```

Both AIC and BIC are **lower** for the reduced model. Since smaller AIC/BIC values indicate a better balance between model fit and model complexity, this provides further evidence that the reduced model is preferred. The reduced model achieves essentially the same explanatory performance as the full model while using fewer predictors, and AIC/BIC reward this parsimony.

Final Conclusion, Literature Connection, and Limitations

Answer to the Research Question

Our research question asked: *How and which financial market indicators drive daily movements in silver returns, affecting silver’s positioning in portfolios and financial markets broadly?*

Based on the final reduced model, the evidence shows that: - **Gold returns** have the strongest positive effect on silver returns, indicating a close and statistically significant co-movement between the two metals. - **Equity market returns (S&P 500 and Nasdaq)** also influence silver returns, suggesting that silver behaves partly as a “risk-on” asset rather than a pure safe-haven. - **Oil returns** significantly affect silver, reflecting the industrial-demand channel for silver. - **Lagged effects from gold and equity markets** provide additional short-term predictive power.

Therefore, the analysis concludes that **silver returns are jointly driven by precious metal dynamics (gold), broad equity market movements, and industrial commodity conditions**, demonstrating that silver is a hybrid asset combining safe-haven and cyclical characteristics.

```
summary(reduced_model)$coefficients
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	-7.501971e-04	6.443365e-04	-1.1642940	2.443806e-01
##	gold_ret	1.465287e+00	1.819666e-02	80.5250727	0.000000e+00
##	sp500_ret	3.758408e-01	4.413878e-02	8.5149792	2.396517e-17
##	nasdaq_ret	-7.076384e-02	3.663305e-02	-1.9316940	5.347425e-02
##	oil_ret	1.995769e-02	4.363943e-03	4.5733160	4.959261e-06
##	gold_ret_lag1	3.225053e-02	1.814963e-02	1.7769248	7.566348e-02
##	sp500_ret_lag1	4.219819e-02	1.645084e-02	2.5651078	1.035381e-02
##	platinum.close	3.808286e-06	5.554327e-06	0.6856431	4.929816e-01

Connection to the Literature

Our empirical findings are highly consistent with the evidence reported in the peer-reviewed literature on silver markets. The strong and highly significant positive effect of gold returns on silver returns directly supports Cohen (2022), who documents short-run cross-metal spillovers from gold to silver and highlights silver as the most predictable precious metal using linear regression models. The positive and significant influence of S&P 500 and oil returns on silver echoes the results of Adamkovičová & Blažek (2021), who find that silver co-moves with equity markets and industrial-demand indicators such as oil, reflecting silver’s hybrid role as both a precious and industrial metal. Consistent with their findings, our model also shows no meaningful impact from FX factors once returns are used.

In addition, the absence of a significant daily effect of platinum in our model aligns with Fatima et al. (2022), who report long-run co-movement between silver and platinum but do not identify a clear short-run directional relationship. Taken together, the regression results reinforce the literature’s broader conclusion: silver exhibits characteristics of both a safe-haven asset (via its tight link to gold) and a cyclical asset sensitive to broader market and economic conditions (via equities and oil).

Limitations and Impact

1. Heavy-Tailed Residuals:

The Q–Q plot shows deviations in the tails, a common feature in financial return data. This reduces the accuracy of extreme-value predictions and explains the wide prediction intervals observed.

2. Mild Heteroskedasticity:

The Scale–Location plot suggests slightly non-constant variance. This may affect the precision of standard errors, but the main coefficient conclusions remain stable.

3. Omitted Variables:

Important indicators such as volatility indices (VIX), interest rates, or macroeconomic announcements are not included. Their omission may leave some short-run variation unexplained.

4. Time-Series Features Not Modeled:

Silver returns may exhibit autocorrelation or regime shifts (e.g., crisis vs non-crisis periods). Our simple OLS model cannot capture such dynamics, which may be an important limitation, limiting medium- and long-horizon predictive accuracy.

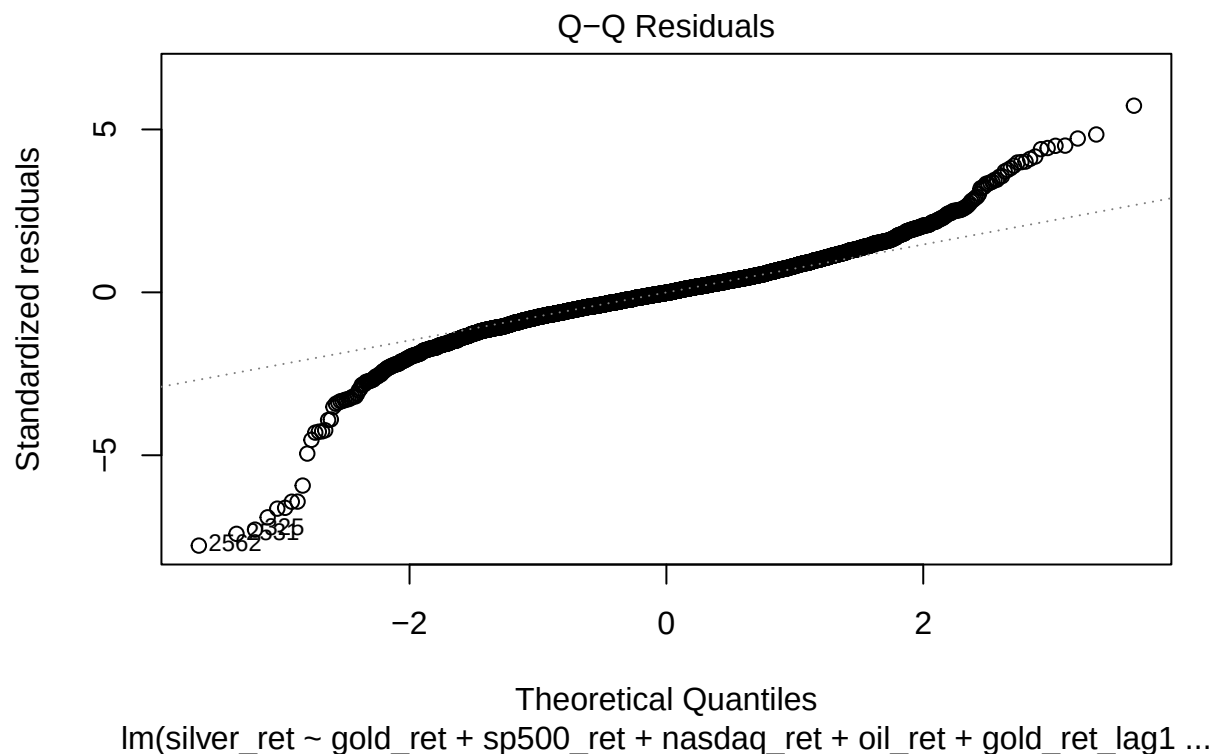
5. Market Microstructure Noise:

Daily closing prices may contain non-synchronous trading effects across assets, which can introduce additional noise into regressions. Although this noise does not invalidate the model, it introduces an unavoidable source of measurement error that should be recognized when interpreting the results.

Despite these limitations, the major relationships - particularly the influence of gold, equities, and oil - remain consistent with theory, diagnostics, and peer-reviewed literature.

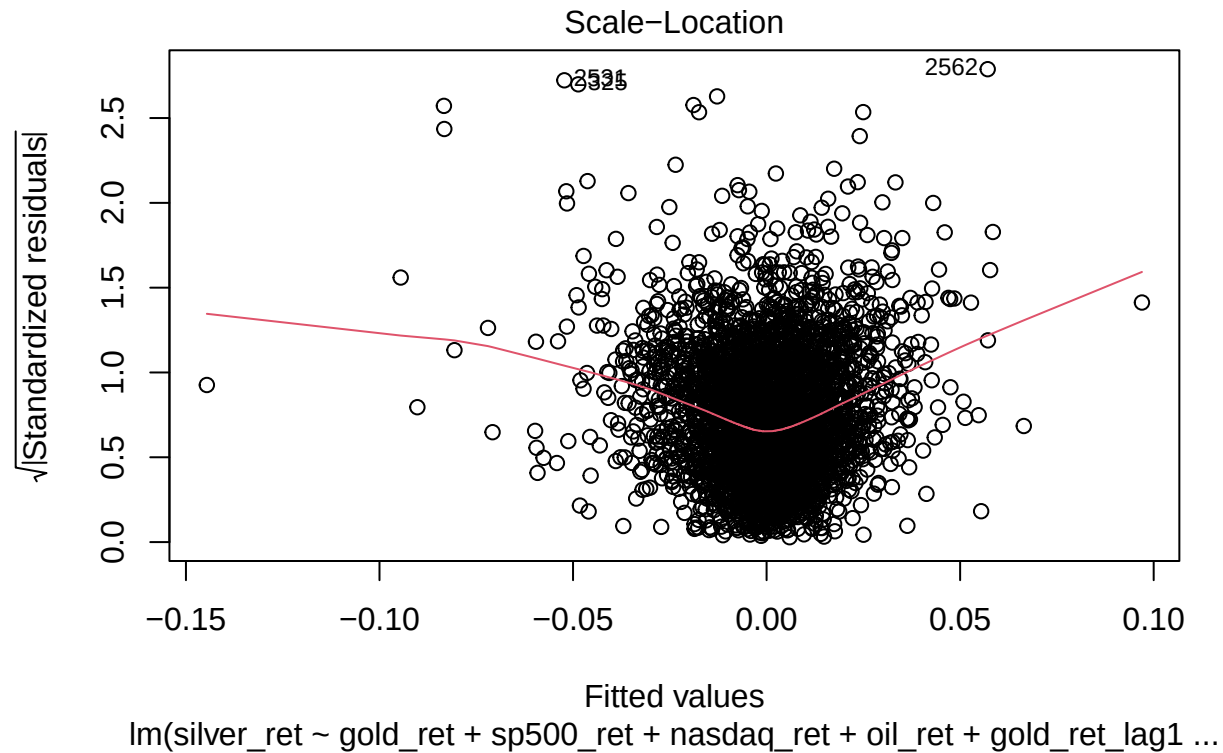
Heavy tails (Q–Q):

```
plot(reduced_model, which = 2)
```



Heteroskedasticity (Scale–Location):

```
plot(reduced_model, which = 3)
```



Overall Conclusion

Our study set out to identify which financial market indicators best explain daily movements in silver returns, and whether these relations justify how silver is perceived in financial markets today, especially given the tremendous year that silver has had, and how “safe” assets have been reimagined. Using a carefully selected and diagnostically validated OLS model, we find consistent evidence that silver’s behavior is shaped by a combination of precious-metal dynamics and broader macro-financial forces. Gold remains the strongest and most reliable predictor of silver, reinforcing silver’s identity as a precious metal with safe-haven features. At the same time, the significant positive effects of S&P 500 and oil returns show that silver responds to global economic activity and risk sentiment, reflecting its substantial industrial use and partial alignment with risk-on market conditions. Short-run spillovers from lagged gold and equity returns further indicate that silver’s daily movements are influenced by immediate market momentum.

Taken together, these results paint a nuanced picture: silver is neither a pure safe-haven asset nor a purely cyclical commodity. Instead, it occupies an intermediate role where both defensive and pro-cyclical forces shape its returns. This conclusion is consistent with findings across the recent literature and underscores the importance of modeling silver as a hybrid asset when evaluating its financial behavior.